# Product Specification – Date Liquid Sugar (Golden Date Syrup)

## 1. Product Information:

| Aspect | Description |
|---|---|
| **Product Name** | Date Liquid Sugar (Golden Date Syrup) |
| **Product Code** | SUFL-7297 |
| **Crop Year** | 2021 |
| **Allergens** | No Allergen. As described in EU Regulation no. 1169/2011 |
| **Product Description and Typical Appearance** | Date sugar is a natural nutrient extracted from additive free date concentrate. Its appearance is completely transparent and the colour is bright to deep yellow. The product processing includes extraction, clarification, concentration, additional clarification and filtration which are not done for normal date syrup. By clarification, Insoluble, Soluble (colorful), Semisoluble (pectin) substances are removed.<br><br>Date sugar contains sugars such as fructose and glucose and a trace of others which might be an alternative to other sweetners especially natural honey. Sweetining level of date fructose is 1.2 times more than sucrose resulting in the weight decrease in its industrial consumption at the same Brix. Therefore it is competitive to other sweetners commercially.  |
| **Applications** | 1. Date liquid sugar can be used in production of important foodstuff groups like natural juices, ice-creams, dairy and bakery products, various beverages and special beverages for athletes.<br>Therefore it can be the best alternative for syrup due to its sugar content.<br>2. Date sugar is the best sweetener for food industries because of its numerous advantages as reasonable price in comparison to similar types like honey as well as better and sweeter taste.<br>3. With cola production increasing trend in the world, it is notable that date sugar production has generated a great revolution in cola production. So, most cola producers in the world have tendency to use date sugar as a sweetening substance.<br>4. Date sugar can be used in the production of isotonic and energetic beverages, bakery products (cake and confectionery products), dairy products, etc. and it is an appropriate alternative to HFCS in cola and juice. Also in ice-cream it helps to create microcrystals because of reducing freezing point.<br>5. Sugar structure of date sugar including fructose and glucose is the same as honey. Date sugar helps to improve pressure of body liquids (due to the low ratio of Na to K). Besides, date sugar has high level of fructose (sugar special for diabetics there is in dried mulberry). Hence, it can be used instead of allowable sugar, diabetic people use since fructose is a type of sugar which its metabolism is independent of insulin. On the other hand date sugar contains high level of potassium while its sodium is lower then it is<br>recommended to the people who are on diet.<br>6. Sugar in date is immediately absorbed and converted to energy after eating. Date is useful for refreshment and obviation of anemia due to high content of iron, fiber and potassium. Date contains various types of vitamin B, being effective in making nervous and digestive systems steady. Date is an appetizer and cleaning agent of liver and kidney. It should be mentioned liquid date sugar supplied by Sahara UK Foods has all of the characteristics of date. |

## 2. Chemical Properties:

| Parameter | Description |
|---|---|
| **Brix** | 68 - 72 |
| **PH** | 3.5 – 4.5 |
| **Acidity** | < 0.1 |
| **Turbidity** | < 10 |
| **Sucrose** | < 2 |
| **Fructose** | 27 - 32 |
| **Glucose** | 32 - 38 |
| **5-HMF** | 5 - 35 |

3. **Nutritional Information Per 100 gr:**

| Items | Value | Items | Value |
|---|---|---|---|
| **Magnesium** | 54 mg | **Calories** | 277 Kcal |
| **Potassium** | 696 mg | **Fat** | 0 |
| **Phosphorus** | 62 mg | **Cholestrol** | 0 |
| **Sodium** | 1 mg | **Total Sugar** | 75 |
| **Zinc** | 0.4 mg | **Sucrose** | 0.53 gr |
| **Copper** | 0.4 mg | **Fructose** | 31.95 gr |
| **Manganese** | 0.3 mg | **Vitamin A** | 149 IU |
| **Protein** | 0 mg | **Vitamin B6** | 0.2 mg |
| **Sulfur** | 0 mg | **Vitamin E & D** | 0 |
| **Humidity** | 21.1 gr | **Other Vitamins** | 16 mg |
| **Ash** | 1.7 gr | **Calcium** | 64 mg |
| | | **Iron** | 0.9 mg |

4. **Comparison of Honey – (White Sugar) Sucrose – Dare Liquid Sugar:**

| Characteristic | Date Liquid Sugar (Per 100g) | Sucrose (White Sugar) (100g) | Honey (Per 100g) |
|---|---|---|---|
| Calories | 277 Kcal | 387 Kcal | 304 Kcal |
| Fat | 0 | 0 | 0 |
| Cholesterol | 0 | 0 | 0 |
| Protein | 0 | 0 | 0 |
| Total Sugar | 75 | 99.91 | 82 |
| Sucrose | 0.53 gr | 99.88 gr | 0.89 gr |
| Glucose | 33.68 gr | 0.3 gr (Invert Sugar) | 35.74 gr |
| Fructose | 31.95 gr | | 40.93 gr |
| Vitamin A | 149 IU | 0 | 0 |
| Vitamin C | 0 | 0 | 0.1 |
| Vitamin B6 | 0.2 mg | 0 | 0 |
| Vitamin E & D | 0 | 0 | 0 |
| Other Vitamins | 16 mg | 0 | 4.1 mg |
| Calcium | 64 mg | 0 | 6 mg |
| Iron | 0.9 mg | 0 | 0.4 mg |
| Magnesium | 54 mg | 0 | 2 mg |
| Potassium | 696 mg | 2 mg | 52 mg |
| Phosphorus | 62 mg | 0 | 4 mg |
| Sodium | 1 mg | 0 | 4 mg |
| Zinc | 0.4 mg | 0 | 0.2 mg |
| Copper | 0.4 mg | 0 | 0 |
| Manganese | 0.3 mg | 0 | 0.1 mg |
| Humidity | 21.3 gr | 0.05 gr | 17.1 gr |
| Ash | 1.7 gr | 0.028 | 0.2 gr |

**Red colour indicates disadvantages of other sweetners.**

**Green colour indicates advantages of Date Sugar to the other sweetners.**

5. **Metal Detection:**

| Criteria | Minimum Detection Size (Sphere size) | Units |
|---|---|---|
| Ferrous | 2.5 | mm |
| Non-Ferrous | Cu: 3.5, Al: 2.5 | mm |
| Stainless Steel | 3 | mm |

6. **Shelf Life and Storage Instructions:**
   - Production Date + 18 months.
   - Keep in cool and dry ambient, store away from sources of strong odors or which may breed insect. Rotate stocks by using oldest first.

7. **Genetically Modified Organisms (GMO):**
   - No Genetically Modified Organisms are permitted.

8. **Fumigation:**
   - Container with goods is fumigated immediately prior to shipment with PH3 gas at a specific dosage and time according to the temperature.

   2 GRAMS PH3 / CUBIC METER FOR 4 DAYS AT 16 °C – 25 °C
   2 GRAMS PH3 / CUBIC METER FOR 3 DAYS AT Over 25 °C

9. **Allergens:**
   - No Allergens as per the EU regulation no. 1169/2011

| **Agreement of Specification** – If no signed copy of this specification is received within 28 days of issue to the Customer then this will be considered accepted. | | | |
|---|---|---|---|
| **Supplier Name** | Sahara UK Foods Limited | **Customer Name** | |
| **Name** | Hassan Syed | **Name** | |
| **Signature (on behalf of SUFL)** | *H.Syed* | **Signature (on behalf of Customer)** | |
| **Position** | Quality Manager | **Position** | |
| **Date** | 17/06/2021 | **Date** | |